How big is the multiplier? What Australian data say about government spending

> The textbook says an extra dollar spent ripples through the economy and adds more than a dollar to income. A 2026 study of Australian data finds the effect depends on the type of spending and on how much slack the economy has.

k = 1 ÷ (1 − MPC)
Simple multiplier formula. If MPC = 0.8, then k = 5

0.94
Government consumption multiplier after six quarters when unemployment is high

−0.95
Government investment multiplier at six quarters (average conditions)

−2.54
Government investment multiplier at six quarters when unemployment is low

81% / 12%
Share of government spending that was consumption / investment, 1984 to 2019

1984-2019
Period of Australian quarterly data studied

1. The textbook multiplier

Government spends $1 → Becomes income for workers and firms → They spend a share (MPC) → That becomes someone else's income → Total rise in GDP is a multiple of $1

The simple multiplier is **k = 1 ÷ (1 − MPC)**, or equivalently 1 ÷ (MPS + MPT + MPM), where the marginal propensities to save, pay tax and import are the withdrawals from each round of spending. Example: MPC of 0.75 gives k = 4, so a $100 million rise in injections lifts national income by up to $400 million. The multiplier is smaller in an open economy because some spending leaks to imports, taxes and saving.

2. What the Australian evidence says (Reynolds and Fisher, 2026)

Zac Reynolds and Lance Fisher (Macquarie University) estimated dollar-for-dollar multipliers for all levels of Australian government using quarterly data from 1984 to 2019. They split the economy into a **high unemployment state** (unemployment above its trend rate) and a **low unemployment state**.

Type of spending	Finding
Government consumption (wages of public servants, goods and services)	Crowds in private activity **only when unemployment is high**, and only moderately: the cumulative multiplier is 0.94 at six quarters. Otherwise the effect is small and not statistically significant.
Government investment (infrastructure, buildings)	Average multiplier of −0.95 at six quarters: it **crowds out** private activity. The crowding out is stronger (−2.54) when unemployment is low, and negligible when unemployment is high.

- **Meaning of crowding out:** in a near full employment economy, extra government demand competes for the same workers and resources, so private spending and investment fall.
- **Limits:** the sample ends in 2019 and leaves out COVID. Estimates are statistical and the authors say they are "robust" to changes in method, but they are not a forecast of any single policy. A multiplier below one means the spending did not add a full dollar to GDP.

Exam angle: link the formula to the evidence. Use k = 1/(1 − MPC) to calculate, then evaluate: real-world multipliers are smaller than the formula suggests because of leakages, and they depend on spare capacity. This supports fiscal policy being more effective when the economy has slack, and is a good point in an evaluate or assess response on discretionary fiscal policy.

Syllabus link

HSC 2009 Topic 3 (3A Economic activity): 3A.3 the simple multiplier. Also supports 3A.2 (injections and withdrawals) and 4C on fiscal policy. The paper is filed on this site.

Try this

1. Calculate the multiplier when MPC is 0.8. (1 mark)
2. Explain why the multiplier in Australia is smaller than the formula predicts. (3 marks)
3. Assess the effectiveness of an increase in government spending when unemployment is high. (6 marks)

Sources: Reynolds, Z. and Fisher, L. A., Estimates of government spending multipliers in Australian data, Applied Economics, 2026, vol. 58, no. 5, pp. 904-920 — https://doi.org/10.1080/00036846.2025.2461326